THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–12) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. Which of these is the best reason for taking repeated measurements? 1

(a) to waste time **(b)** to make the object longer
(c) to reduce random errors **(d)** to change units

2. A bottle cap made of plastic is replaced by a metal cap. Which problem might arise? 1

(a) it floats **(b)** it becomes transparent
(c) it may rust or conduct heat **(d)** it dissolves

3. Which of the following devices uses an insulator to protect the user? 1

(a) electric iron handle **(b)** kettle body
(c) iron rod **(d)** frying pan base

4. Which of the following changes is reversible? 1

(a) melting of ice **(b)** rusting **(c)** cooking of rice **(d)** burning of wood

5. Which of the following correctly orders the steps of a municipal water treatment? 1

(a) sedimentation, filtration, disinfection **(b)** boiling, magnet
(c) disinfection, filtration **(d)** filtration, sedimentation

6. Which characteristic makes 'excretion' essential for life? 1

(a) it makes food **(b)** it helps in growth
(c) accumulated wastes are toxic **(d)** it makes light

7. A village uses a solar pump instead of a diesel pump. The main benefit is: 1

(a) more fuel **(b)** no fuel cost and no smoke
(c) more smoke **(d)** more noise

8. How would the length of day change if the Earth rotated twice as fast? 1

(a) halve **(b)** double **(c)** stay the same **(d)** vanish

9. Which of these is the best example of a testable question? 1

(a) Which sport is exciting?
(b) Does temperature affect how fast sugar dissolves?
(c) Is blue a better colour than red?
(d) Is a mango tastier than a pear?

10. Which group of features suits an animal living in the polar region? 1

(a) feathers, hollow bones, beak **(b)** gills, fins, scales
(c) thick fur, fat layer, white colour **(d)** thin skin, long legs, hump

11. Which pair correctly matches the deficiency and disease? 1

(a) iodine, anaemia **(b)** vitamin A, night blindness
(c) iron, scurvy **(d)** vitamin C, rickets

12. A compass kept on a table shows north in a wrong direction. The best explanation is: 1

(a) the compass is in sunlight **(b)** the table is made of wood
(c) the compass needle is too light **(d)** a magnet or iron object nearby

13. A thread of length 188 cm goes exactly 4 times round the edge of a circular plate. The circumference of the plate is: 1

(a) 752 cm **(b)** 47 cm **(c)** 94 cm **(d)** 188 cm

14. Which of these shows that solubility depends on the substance and not only on appearance? 1

(a) sugar and chalk powder both look white but behave differently in water
(b) both are white
(c) both are powders
(d) both are solid

15. A student records water temperature in the evening: 20 °C, 20 °C, 20 °C after placing in a room at 20 °C. This shows: 1

(a) the water is cooling
(b) the water is boiling
(c) the water is at the same temperature as the room
(d) the water is being heated

16. If the surface area of a liquid is doubled, the rate of evaporation will: 1

(a) increase **(b)** stay the same **(c)** stop **(d)** decrease

17. A mixture of two liquids that do not mix is separated by: 1

(a) winnowing **(b)** filtration through paper
(c) decantation after the layers form **(d)** sieving

18. Which of these cannot be taken as proof of life when used alone? 1

(a) respiration **(b)** growth from inside
(c) reproduction **(d)** movement

19. A hill slope is turned into steps and trees planted. The main effect is: 1

(a) higher hill **(b)** more soil lost
(c) no rain **(d)** less soil washed away

20. On a full-moon night the Moon rises roughly at: 1

(a) sunrise **(b)** noon **(c)** sunset **(d)** midnight

21. C.V. Raman's work on the scattering of light began with his curiosity about why the sea looks blue. This shows that: 1

(a) curiosity about everyday things can lead to great discoveries
(b) experiments are unnecessary in science
(c) only costly equipment can lead to discoveries
(d) great discoveries need no observation

22. The roots of a banyan hanging from branches are called: 1

(a) prop roots **(b)** fibrous roots **(c)** tap roots **(d)** tuberous roots

23. Which food pair shows foods rich in fat and protein respectively? 1

(a) salt and sugar **(b)** rice and butter **(c)** butter and dal **(d)** apple and orange

24. A pin stuck to a magnet can also pick up another pin because: 1

(a) the pin is sticky **(b)** pins have glue
(c) air presses them **(d)** the pin acts as a temporary magnet

25. Which statement about a ceiling fan switched on is correct? 1

(a) the blades move randomly **(b)** each blade moves linearly
(c) the fan is at rest **(d)** each blade moves in a circular path

26. After removing iron filings, adding water to the mixture and filtering separates: 1

(a) sand on the filter paper **(b)** salt on the filter paper
(c) both on the filter **(d)** nothing

27. Which temperature is not possible on a clinical thermometer scale? 1

(a) 20 °C **(b)** 40 °C **(c)** 36 °C **(d)** 42 °C

28. A tap leaks 1 drop per second; if 20 drops make 1 mL, the leak in one hour is about: 1

(a) 18 mL **(b)** 180 mL **(c)** 3600 mL **(d)** 60 mL

29. A mixture of sand and iron in 1:1 by mass weighing 40 g. After using a magnet, the iron obtained should be: 1

(a) 20 g **(b)** 0 g **(c)** 40 g **(d)** 10 g

30. A calf grows into a cow. Its growth involves: 1

(a) melting **(b)** growth from within using food
(c) absorption of sand **(d)** addition of stones

31. A farmer notes that crop yield fell year after year in the same field with the same crop. Probable reason: 1

(a) nutrient loss in soil **(b)** more worms
(c) more sunshine **(d)** more rain

32. A satellite completes 16 orbits in one day (24 hours). The time taken for one orbit is: 1

(a) 90 minutes **(b)** 24 hours **(c)** 60 minutes **(d)** 120 minutes

33. In a graph of plant height against days, a steady upward line shows: 1

(a) the plant is not changing **(b)** the plant is shrinking
(c) the soil is dry **(d)** the plant grows with time

34. The most likely reason that a cactus has thick, fleshy stem is: 1

(a) to hold soil **(b)** to make food in the dark
(c) to attract birds **(d)** to store water

35. A mother wishes to improve her child's diet cheaply. Best advice: 1

(a) give only sweets
(b) give only milk
(c) include pulses, seasonal vegetables and local fruits
(d) buy imported snacks

36. Which of these is the correct way to hold a compass while taking a reading? 1

(a) flat on the palm, away from iron objects **(b)** close to a speaker
(c) tilted near a steel desk **(d)** in a metal box

37. Parallax error is least when the scale is: 1

(a) tilted
(b) viewed from the side
(c) placed far away
(d) placed touching the object and viewed straight

38. Which of these should not be done with glass items? 1

(a) washing with water **(b)** drying
(c) keeping on a shelf **(d)** heating suddenly

39. The difference between heat and temperature, as at this level, is that temperature tells: 1

(a) how hot something is **(b)** how much it weighs
(c) its colour **(d)** its volume

40. A closed bottle with water and air is heated. What happens to the vapour? 1

(a) it freezes **(b)** it increases **(c)** it turns to ice **(d)** it vanishes

41. Which is a reason for decanting slowly? 1

(a) to avoid disturbing the sediment **(b)** to cool
(c) to dissolve **(d)** to save time

42. Which organ in a plant takes in carbon dioxide for food making? 1

(a) seed **(b)** leaf **(c)** root **(d)** flower

43. Why would a sunny region benefit from solar panels? 1

(a) sunlight pollutes the air
(b) renewable energy is available there all year
(c) sunlight is limited there
(d) panels need no sunlight

44. Which two planets can be seen crossing the Sun's disc from Earth? 1

(a) Saturn and Neptune **(b)** Mars and Jupiter
(c) Mercury and Venus **(d)** Uranus and Pluto

45. A farmer sees insects on her crop and uses neem spray on half of the field. What is the control in this test? 1

(a) the sprayed half **(b)** the neem tree
(c) the half field not sprayed **(d)** the neighbours' field

46. A fish placed on land dies quickly because: 1

(a) it needs a lot of salt **(b)** its gills cannot take oxygen from air
(c) it is too bright **(d)** it is too cold

47. Why can a diet only of rice cause weakness even though it has plenty of energy? 1

(a) it has too much fibre **(b)** it lacks protein and many vitamins
(c) it contains poison **(d)** it has no starch

48. An iron nail dangling from a magnet picks up another nail. The second nail is held because: 1

(a) gravity
(b) glue is present
(c) the first nail became temporarily magnetised
(d) air presses

49. The thickness of 300 sheets of paper is 1.5 cm. The thickness of one sheet in mm is: 1

(a) 0.005 mm **(b)** 0.5 mm **(c)** 0.05 mm **(d)** 5 mm

50. Which statement is correct? 1

(a) a material that is opaque may be shiny **(b)** all shiny materials are transparent
(c) all opaque materials are dull **(d)** all transparent materials are soluble

END OF QUESTION PAPER **Total: 50 Marks**

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